23IT601	PRINCIPLES OF COMMUNICATION SYSTEMS	3/0/0/3
Nature of Course	C (Theory Concept)	
Pre requisites	-	
Course Objectives:		
1.	To develop a comprehensive understanding of communication systems and their fundamental characteristics	
2.	To introduce various modulation techniques, including AM, FM, and PM, and their applications	
3.	To explain the concepts of random processes, noise, and their influence on digital transmission	
4.	To explore information theory, entropy, source coding, and error control techniques	
5.	To acquire the knowledge about the principles of transmission media and multiplexing methods like TDM, FDM, and CDM	
Course Outcomes		
Upon completion of the course, students shall have ability to		
C601.1	Demonstrate an understanding of modulation techniques and their significance in communication systems	[AP]
C601.2	Apply various noise models and analyze their effects on communication systems	[AP]
C601.3	Analyze source coding and error correction techniques for reliable digital transmission	[AN]
C601.4	Distinguish between different multiplexing methods and their applications in wired and wireless media	[AN]
C601.5	Utilize Shannon's Theorem to evaluate communication system performance	[AP]

INTRODUCTION TO COMMUNICATION SYSTEMS AND ANGLE MODULATION
15 Hours

Overview of Communication Systems – Channels and their characteristics-Modulation - Benefits-Generation and demodulation of AM-DSBSC, SSB and VSB Signals-Comparison of Amplitude Modulation Systems-Frequency Translation-Non -Linear Distortion-Phase and frequency modulation-Single Tone-Narrow band-wideband FM-Transmission Bandwidth-Generation and demodulation of FM Signal

RANDOM PROCESSES, NOISE, AND DIGITAL TRANSMISSION **15 Hours**

Overview of probability -Random Processes-Gaussian Process-Shot Noise-Thermal noise and white noise-Narrow band noise -Noise Margin-Noise temperature -Noise Figure - Pulse Modulation Schemes: Pulse Amplitude Modulation (PAM) - Pulse Width Modulation (PWM) - Pulse Position Modulation (PPM)-Pulse Code Modulation (PCM)- Amplitude Shift Keying (ASK), Frequency Shift Keying (FSK)

INFORMATION THEORY & TRANSMISSION MEDIA **15 Hours**

Entropy and information-Mutual information and Shannon's Theorem-Source coding techniques: Huffman coding, Shannon-Fano coding-Error Control Coding-Hamming codes, cyclic redundancy check (CRC)-Block codes, convolutional codes-Forward error correction (FEC)-Transmission Media & Multiplexing-Wired & wireless media-Time-Division Multiplexing (TDM)-Frequency-Division Multiplexing (FDM)-Code-Division Multiplexing (CDM)

Total Hours	**45**

Text Books:	
1.	Taub, H. Schilling, D.L, Saha G, “Principles of communication systems” McGraw - Hill Higher education ,4th Edition, 2017.
2.	Lathi B.P & Zhi Ding, H. M. Gupta "Modern Digital and Analog Communication Systems," Oxford University Press, 4th Edition, 2017.
Reference Books:	
1.	Leis J.W, “Communication system principles using MATLAB”, John Wiley & sons, 1st Edition, 2018
2.	Taub & Schilling, "Principles of Communication Systems," McGraw-Hill, 4th Edition, 2017.
Web References:	
1.	https://www.tutorialspoint.com/principles_of_communication/index.htm
2.	https://www.ee.columbia.edu/~egc/notes.pdf?
Online Resources:	
1.	https://archive.nptel.ac.in/courses/108/104/108104091/
2.	https://www.udemy.com/course/learn-analog-communication-from-basics/
3.	https://www.classcentral.com/course/swayam-principles-of-communication-systems-i-7963